Assignments-2-PC 505 AI-Machine Learning

Unit2

1. Describe Perceptron and any two activation functions. What are the limitations of perceptron.
2. Briefly describe the various layers of Multilayer perceptron.
3. Describe the forward propagation and loss functions in MLP.
4. Briefly describe the perceptron algorithm.

Unit3

1. What is support vector? Explain how Support Vector Machine works with suitable examples
2. What are SVM Kernels. Write notes on (i) Linear Kernel (ii) Polynomial Kernel (iii) Radial basis function kernel (iv) Sigmoid kernel
3. What is the role of Bayesian models in Machine learning.
4. Define Naïve Bayes Classifier and describe Naïve Bayes Algorithm.
5. What is Overfitting? Briefly explain under what circumstances the generalization performance increases due to overfitting.
6. Write note on
   (i) Hidden Markov Model
   (ii) Bayesian Network
7. a) Differentiate between Discriminative and Generative learning
   b) State Bayes Theorem
   c) Consider a coin that is known to be either fair or 60% in favour of heads. Determine which kind of coin it is based on the observations, given
   $P(h_1) = 0.75$ and $P(h_2) = 0.25$

Unit4

1. A study is conducted involving 10 patients to investigate the relationship and effects of patients age and the blood pressure. Calculate the linear regression of patients age and blood pressure

	1	2	3	4	5	6	7	8	9	10
Age	35	40	38	44	67	64	59	69	25	50
BP	112	128	130	138	158	162	140	175	125	142

2. Write notes on
   (i) Feed forward Neural Networks
   (ii) Back Propagation
   (iii) Convolution Neural Networks
   (iv) LeNet 5 Architecture
   (v) Bagging
3. Write notes on
   (i) Multilayer Perceptron
   (ii) Neural Network Architecture
   (iii) Back Propagation Algorithm
4. Write notes on
   (i) Regularization
   (ii) K-Fold cross Validation

Unit5	1. Compute the Linear Discriminant projection for the following dimensional data set Samples for class w_1:X_1={(4,2), (2,4), (3,6), (4,4)} Samples for class w_2:X_2={(9,10), (6,8), (9.5), (8,7), (10,8)} 2. Compute the Linear Discriminant projection for the following dimensional data set Samples for class w_1:X_1={(1,2), (2,3), (3,3), (4,5), (5,5)} Samples for class w_2:X_2={(4,2), (5,0), (5,2), (3,2), (5,3), (6,3)} 3. Briefly describe the steps to calculate a lower dimensional subspace of the LDA technique. 4. What is Reinforcement Learning? What are the elements of Reinforcement Learning? What is the scope and limitations of Reinforcement Learning. 5. Write shot notes on (i) LDA (ii) PCA (iii) K-means Algorithm (iv) Hierarchical Clustering (v) Clustering